

切記：計算過程比答案更重要，沒有計算過程不給分

Total Mark : 100 pts

■ 1 (10pts). 試求 $\sin \frac{\pi}{3} \cos \frac{\pi}{6} + \tan \frac{\pi}{4} \cot \left(-\frac{\pi}{4}\right) + \sin \left(-\frac{11\pi}{6}\right) \cos \frac{\pi}{3}$ 之值.

(練習題本 Sec.4 #1)

■ 2 (10pts). 已知 x 、 y 為實數且 $3x + 4y = 10$ ，試求 $x^2 + y^2$ 的最小值.

(P.63 練習 16)

■ 3 (15pts). 設 $A(-13, -19)$ 、 $B(x, y)$ 為平面上相異兩點，若 $\overrightarrow{AB}$ 與 $\vec{u} = (5, 12)$ 同方向且 $|\overrightarrow{AB}| = 26$ ，試求 $3x - 4y$ 之值.

(P.65 統測常考題—例題 2)

■ 4 (15pts). 設平面上三點 $A(1, 1)$ 、 $B(5, -2)$ 、 $C(5, 2)$ ，且 $\overrightarrow{AC}$ 在 $\overrightarrow{AB}$ 的正射影為 $\overrightarrow{AD}$ ，若 $\overrightarrow{DC} = (x, y)$ ，求 $x + y$ 的值.

(P.69 #13)

■ 5 (15pts). 若 $\triangle ABC$ 的三邊長為 4、5、6，則其外接圓直徑為何？

【Hint：海龍公式與三角形面積公式】

(P.49 #14)

■ 6 (35pts). 設 θ 為銳角，若 $\sin \theta + \cos \theta = \frac{4}{3}$ ，試求下列各題之值：

(a) (15pts) $\sin \theta - \cos \theta$.(b) (10pts) $\tan \theta + \cot \theta$.(c) (10pts) $\sec \theta + \csc \theta$.【Hint：想看看須先算出 這格不計分不用回答，只是提示你而已，才能求這三問答案】

(P.33 邁向統測之路—老師講解(改))

-----答案卷未寫上姓名學號扣總分 10 分，無須繳回題目卷-----

Solution:

$$\begin{aligned}\#1 \quad & \sin \frac{\pi}{3} \cos \frac{\pi}{6} + \tan \frac{\pi}{4} \cot \left(-\frac{\pi}{4}\right) + \sin \left(-\frac{11\pi}{6}\right) \cos \frac{\pi}{3} \\ &= \sin \frac{\pi}{3} \cos \frac{\pi}{6} + \tan \frac{\pi}{4} \left(-\cot \frac{\pi}{4}\right) + \sin \frac{\pi}{6} \cos \frac{\pi}{3} \\ &= \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} + 1 \cdot (-1) + \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4} - 1 + \frac{1}{4} = \underline{0} \quad \text{///}\end{aligned}$$

#2 Remark: 平方和 & 兩兩相乘 $\Rightarrow$ 柯西不等式

$$\begin{aligned}\therefore (x^2 + y^2)(3^2 + 4^2) &\geq (3x + 4y)^2 \Rightarrow (x^2 + y^2) \cdot 25 \geq 100 \\ \Rightarrow x^2 + y^2 &\geq 4 \quad \therefore \min = \underline{4} \quad \text{///}\end{aligned}$$

$$\#3 \quad \vec{AB} = B - A = (x - (-13), y - (-19)) = (x + 13, y + 19).$$

$$\therefore |\vec{u}| = \sqrt{5^2 + 12^2} = 13 = \frac{1}{2} |\vec{AB}| \quad \text{且 } \vec{AB} \text{ 和 } \vec{u} \text{ 同向}$$

$$\therefore \vec{AB} = 2\vec{u} \Rightarrow \vec{AB} \parallel \vec{u} \quad (\text{平行分量成比例})$$

$$\therefore \frac{x+13}{5} = \frac{y+19}{12} = 2 \Rightarrow \begin{cases} x+13=10 \\ y+19=24 \end{cases} \Rightarrow \begin{cases} x=-3 \\ y=5 \end{cases}$$

$$\therefore 3x - 4y = 3(-3) - 4 \cdot 5 = -9 - 20 = \underline{-29} \quad \text{///}$$

$$\#4 \quad \vec{AC} = C - A = (5 - 1, 2 - 1) = (4, 1), \quad \vec{AB} = B - A = (5 - 1, -2 - 1) = (4, -3)$$

$$\vec{AD} = \frac{\vec{AC} \cdot \vec{AB}}{|\vec{AB}|^2} \cdot \vec{AB} = \frac{(4, 1) \cdot (4, -3)}{\sqrt{4^2 + (-3)^2}^2} (4, -3) = \frac{13}{25} (4, -3) = \left(\frac{52}{25}, -\frac{39}{25}\right)$$

$$\therefore \vec{DC} = \vec{DA} + \vec{AC} = -\vec{AD} + \vec{AC} = \left(-\frac{52}{25}, \frac{39}{25}\right) + (4, 1) = \left(\frac{48}{25}, \frac{64}{25}\right) \Rightarrow x + y = \underline{\frac{112}{25}} \quad \text{///}$$

#5 method 1: 海龍公式

$$S = \frac{4+5+6}{2} = \frac{15}{2}$$

$$\Delta ABC = \frac{abc}{4R}$$

$$\therefore \Delta ABC = \sqrt{\frac{15}{2} \cdot \left(\frac{15}{2} - 4\right) \left(\frac{15}{2} - 5\right) \left(\frac{15}{2} - 6\right)} = \sqrt{\frac{15}{2} \cdot \frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2}} = \frac{15\sqrt{7}}{4} = \frac{4 \cdot 5 \cdot 6}{4R}$$

$$\Rightarrow R = \frac{4 \cdot 5 \cdot 6}{15\sqrt{7}} \Rightarrow 2R = \frac{2 \cdot 4 \cdot 5 \cdot 6^2}{15\sqrt{7}} = \underline{\underline{\frac{16\sqrt{7}}{7}}}$$

$$\#6 \sin\theta + \cos\theta = \frac{4}{3} \Rightarrow \sin^2\theta + \underbrace{2\sin\theta\cos\theta}_{=1} + \cos^2\theta = \frac{16}{9} \Rightarrow 1 + 2\sin\theta\cos\theta = \frac{16}{9}$$

$$\Rightarrow \sin\theta\cos\theta = \frac{7}{18}$$

$$a) (\sin\theta - \cos\theta)^2 = \sin^2\theta - 2\sin\theta\cos\theta + \cos^2\theta = 1 - 2\sin\theta\cos\theta$$

$$= 1 - 2 \cdot \frac{7}{18} = \frac{2}{9}$$

$$\Rightarrow \sin\theta - \cos\theta = \pm \frac{\sqrt{2}}{3} //$$

Remark: $0^\circ < \theta < 45^\circ$ 時 $\sin\theta > \cos\theta \Rightarrow \sin\theta - \cos\theta > 0$

$45^\circ < \theta < 90^\circ$ 時 $\sin\theta < \cos\theta \Rightarrow \sin\theta - \cos\theta < 0$.

因題目只有說銳角, 未明示在哪一個區間, 故兩項皆有可能。

$$b) \tan\theta + \cot\theta = \frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta} = \frac{\sin^2\theta + \cos^2\theta}{\sin\theta\cos\theta} = \frac{1}{\sin\theta\cos\theta} = \frac{1}{7/18} = \underline{\underline{\frac{18}{7}}}$$

$$c) \sec\theta + \csc\theta = \frac{1}{\cos\theta} + \frac{1}{\sin\theta} = \frac{\sin\theta + \cos\theta}{\sin\theta\cos\theta} = \frac{4/3}{7/18} = \frac{4 \times 18^6}{3 \times 7} = \underline{\underline{\frac{24}{7}}}$$